

delayed the project as all resources were now diverted towards the war effort. By 1919, passenger traffic in Bombay had risen to numbers beyond the capacity of steam locomotives. Finally, the Bombay Municipal Corporation (BMC) issued a resolution for the GIPR to electrify its line till the suburb of Kalyan.

The Government drew up schemes for electrification not only in Bombay, but also in the Madras Presidency. All electric railway systems at this time operated on a 1,500-volts direct current electric traction. While the suburban services used Electric Multiple Units (EMUs), the main line was worked by electric locomotives.

Of suburbs ...

A section of the Harbour Line was the first to be electrified when an electric train ran from VT to Kurla on 3 February 1925. The part between Kurla and the Reay Road station had opened in 1910. Further extension of the line could only be on an elevated track due to shortage of land. The hauling power of steam locomotives was insufficient for a steep climb and so the line was chosen to be electrified. In 1926, electrification was carried out on the line from Kurla to Thane and the Mahim Chord on the Harbour Line. By 1936, all lines till the extreme suburb of Virar were electrified.

... and ghats

Traction in hilly sections also resulted in extreme wear and tear for steam locomotives. By 1930, therefore, electrification was introduced on the main line from Kalyan to Igatpuri on the Thul Ghat and Kalyan to Poona (Pune) on the Bhore Ghat (see [Conquering The Ghats](#)). According to Indian author JN Sahni in his book *Indian Railways: One Hundred Years 1853–1953*, passenger trains were worked by special Ghats engines. Goods trains were broken into two parts before they could be transported over the Ghats. Electrification along this route helped in removing the “bottlenecks” on the Ghats.

By 1931, in order to cater to the ever-increasing traffic in Madras City and its southern suburbs, electric power was introduced on a suburban section from Madras Beach to Tambaram. Electrification of this section entailed building double tracks as its single line was ill-equipped to